

Solutions for Midterm 2 Practice Exam

November 2025

Problem 1

- **Answer:** No.

- **Reason 1:** The set does not contain the zero vector.

Consider $(x, y)^\top = (0, 0)^\top$. Then $x - y = 0 - 0 = 0 \neq 1$, so the zero vector is not in the set.

- **Reason 2:** The set is not closed under addition.

Suppose we take two arbitrary vectors $(x_1, y_1), (x_2, y_2) \in V$, which implies $x_1 - y_1 = 1$ and $x_2 - y_2 = 1$. Then:

$$(x_1 + x_2) - (y_1 + y_2) = (x_1 - y_1) + (x_2 - y_2) = 1 + 1 = 2 \neq 1$$

Therefore, the sum of two vectors in V does not necessarily belong to V .

- **Reason 3:** The set is not closed under scalar multiplication.

Let $(x_1, y_1) \in V$, so $x_1 - y_1 = 1$. For any scalar $\alpha \in \mathbb{R}$, we have:

$$\alpha x_1 - \alpha y_1 = \alpha(x_1 - y_1) = \alpha$$

This equals 1 only when $\alpha = 1$, so $(\alpha x_1, \alpha y_1) \notin V$ in general.

Problem 2

- **Answer:** Yes, $x_1 = 1$ and $x_2 = 1$.

- **Explanation:** We verify whether $\vec{v}_3$ can be expressed as a linear combination of $\vec{v}_1$ and $\vec{v}_2$ by finding scalars $x_1, x_2 \in \mathbb{R}$ such that:

$$\begin{aligned}\vec{v}_3 &= x_1 \vec{v}_1 + x_2 \vec{v}_2 \\ &= x_1 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\end{aligned}$$

To solve for x_1 and x_2 , we construct the augmented matrix:

$$\left[\begin{array}{cc|c} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 0 & 1 & 1 \end{array} \right]$$

Applying Gauss-Jordan elimination:

$$\begin{aligned} \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 0 & 1 & 1 \end{array} \right] &\sim \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{array} \right] \\ &\sim \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{array} \right] \end{aligned}$$

From the row-reduced echelon form, we find $x_1 = 1$ and $x_2 = 1$. Therefore, $\vec{v}_3$ can be written as a linear combination of $\vec{v}_1$ and $\vec{v}_2$, which means it lies in the span of $\{\vec{v}_1, \vec{v}_2\}$.

Problem 3

- **Answer:** Yes.
- **Explanation:** For $\mathbf{b}$ to be in $\text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$, it must be expressible as a linear combination of $\mathbf{u}_1$, $\mathbf{u}_2$, and $\mathbf{u}_3$:

$$\mathbf{b} = x_1\mathbf{u}_1 + x_2\mathbf{u}_2 + x_3\mathbf{u}_3,$$

where $x_1, x_2, x_3 \in \mathbb{R}$.

This can be written as a matrix-vector equation:

$$\mathbf{b} = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \mathbf{u}_3] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

To determine whether a solution exists, we can form the augmented matrix and perform row reduction:

$$\begin{aligned} [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \mathbf{u}_3 \mid \mathbf{b}] &= \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 3 \\ 1 & 1 & 2 & 5 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 3 \\ 0 & 1 & 1 & 3 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right] \end{aligned}$$

Since the rank of the coefficient matrix and the augmented matrix are both 2 (i.e., the number of leading 1s or pivots), the system is consistent and has at least one solution. Therefore, $\mathbf{b}$ lies in the span of $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$.

Problem 4

- **Answer:** No.
- **Explanation:** A linear combination of the three vectors in S is given by:

$$x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} x_1 + x_3 \\ x_2 + x_3 \\ 0 \end{bmatrix},$$

where $x_1, x_2, x_3 \in \mathbb{R}$.

Since the third component of every linear combination is always zero, the span of these vectors lies entirely within the xy -plane of $\mathbb{R}^3$. In other words, they can only generate vectors of the form $\begin{bmatrix} a \\ b \\ 0 \end{bmatrix}$, which is strictly a subspace of $\mathbb{R}^3$.

Therefore, the set S does not span $\mathbb{R}^3$.

Problem 5

- **Answer:** $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$
- **Explanation:** $(1, 1, 0)$ can be obtained by

$$1 \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 1 \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Hence, we should remove $(1, 1, 0)$.

The remaining vectors constitute a basis since the only possible solution for

$$x_1 \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x_2 \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_3 \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

is $x_1 = x_2 = x_3 = 0$.

Problem 6

Answer: The inverse of a matrix can be computed using the formula,

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \cdot \text{adj}(\mathbf{A}),$$

where $\text{adj}(\mathbf{A})$ is the adjugate of $\mathbf{A}$. Therefore, if $\det(\mathbf{A}) = 0$, the formula becomes undefined due to division by zero, and the inverse matrix does not exist.

Problem 7

Answer: $\det(\mathbf{A})$ is still 0. The determinant of a square matrix is zero if and only if its rows (or columns) are linearly dependent.

To make the determinant non-zero, the matrix must have a set of linearly independent rows (or columns). This can be achieved by removing dependent rows and replacing them with new rows that are linearly independent of the remaining ones.

Therefore, replacing one row with the sum of the other two rows does not help, since such a row is still linearly dependent on the others. As a result, the determinant remains zero.

Problem 8

Note: Depending on the context or formatting style, a vector may be denoted as either $\vec{a}$ or $\mathbf{a}$.

- **Answer:** $(2, 0, 2)^\top$
- Let $\text{Proj}_{\mathbf{a}}^{\mathbf{w}}$ be the projection vector of $\mathbf{w}$ onto $\mathbf{a}$.
Solving for $\text{Proj}_{\mathbf{a}}^{\mathbf{w}}$:

$$\text{Proj}_{\mathbf{a}}^{\mathbf{w}} = \frac{\mathbf{a} \cdot \mathbf{w}}{\mathbf{a} \cdot \mathbf{a}} \cdot \mathbf{a} = \frac{1 \cdot 1 + 0 \cdot 2 + 1 \cdot 3}{1 \cdot 1 + 0 \cdot 0 + 1 \cdot 1} \cdot (1, 0, 1)^\top = \frac{4}{2} \cdot (1, 0, 1)^\top = (2, 0, 2)^\top$$

Problem 9

The model can be expressed in matrix-vector form as:

$$\vec{y} = \mathbf{X} \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix},$$

where

$$\mathbf{X} = \begin{bmatrix} 3 & 5 \\ 1 & 2 \\ 4 & 6 \end{bmatrix}, \quad \vec{y} = \begin{bmatrix} 10 \\ 5 \\ 12 \end{bmatrix}$$

Therefore, the coefficient matrix, $\mathbf{X}$ is:

$$\mathbf{X} = \begin{bmatrix} 3 & 5 \\ 1 & 2 \\ 4 & 6 \end{bmatrix}$$

Problem 10

- **Answer:** No, it is the global minimum.

- **Critical Point:**

$$\begin{aligned}\nabla f(x, y) &= \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix} \\ &= \begin{bmatrix} \frac{d}{dx}(x^2 + xy) \\ \frac{d}{dy}(y^2 + xy) \end{bmatrix} \\ &= \begin{bmatrix} 2x + y \\ x + 2y \end{bmatrix}\end{aligned}$$

Setting the gradient equal to zero:

$$\nabla f(x, y) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} 2x + y = 0 \\ x + 2y = 0 \end{cases} \Rightarrow (x^*, y^*) = (0, 0)$$

Therefore, the only critical point is at $(0, 0)$.

- **Property of the critical point $(0, 0)$:**

$$D = \frac{\partial^2 f}{\partial x^2} \times \frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 = \frac{\partial(2x + y)}{\partial x} \times \frac{\partial(x + 2y)}{\partial y} - \left(\frac{\partial(2x + y)}{\partial y} \right)^2 = 2 \times 2 - 1^2 = 3$$

Since $D = 3 > 0$ and $\partial^2 f / \partial x^2 = 2 > 0$ everywhere, the critical point at $(0, 0)$ is the global minimum, not a saddle point.

Problem 11

- **Answer:** $(4, -4, 8)^\top, 0$
- **Direction in which the potential increases the fastest**

The function value changes the fastest in the direction of gradient vector at all points. Hence,

we need to first compute the value of gradient vector at $(1, -2, 1)^\top$.

$$\begin{aligned}\nabla U(x, y, z) &= \begin{bmatrix} \frac{\partial U}{\partial x} \\ \frac{\partial U}{\partial y} \\ \frac{\partial U}{\partial z} \end{bmatrix} \\ &= \begin{bmatrix} \frac{d}{dx} 2x^2 \\ \frac{d}{dy} y^2 \\ \frac{d}{dz} 4z^2 \end{bmatrix} \\ &= \begin{bmatrix} 4x \\ 2y \\ 8z \end{bmatrix}\end{aligned}$$

Gradient at $(1, -2, 1)^\top$:

$$\nabla U(1, -2, 1) = \begin{bmatrix} 4 \\ -4 \\ 8 \end{bmatrix}$$

- **Rate of increase in the direction of vector $\mathbf{v} = (1, 1, 0)^\top$**

$$\nabla U(1, -2, 1) \cdot \mathbf{v} = (4, -4, 8) \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}^\top = 0$$

Problem 12

- **Answer:** 9
- The problem reduces to the following definite integral:

$$\int_0^3 x^2 dx = \left[\frac{1}{3} x^3 \right]_0^3 = \left(\frac{1}{3} \cdot 3^3 \right) - \left(\frac{1}{3} \cdot 0^3 \right) = 9 - 0 = 9,$$